

Erdős Problem #1016

EP-1016

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $h(n)$ be minimal such that there is a graph on n vertices with $n + h(n)$ edges which contains a cycle on k vertices, for all $3 \leq k \leq n$. Estimate $h(n)$. In particular, is it true that $h(n) \geq \log_2 n + \log_* n - O(1)$, where $\log_* n$ is the iterated logarithmic function?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>